Martin Fankhauser
Bocconi University
Milan, Italy
martin.fankhauser@unibocconi.it
martin-fankhauser.at

Dear Professor Kock and Members of the Selection Committee,

I am writing to apply for the Postdoctoral Research Fellow position in the Department of Economics at the University of Oxford associated with the ERC project *High-Dimensional Hypothesis Testing*. I am currently completing my PhD in Economics at Bocconi University. I am an econometrician working on partial identification, robust inference, and Bayesian econometrics.

My job market paper, *Sharp Identification for Regressions with Interval-Observed and Missing Covariates*, joint with Gil Peled, develops identification, estimation, and inference for conditional moment models with incompletely observed outcomes or covariates. Without imposing restrictions such as missing-at-random, we characterize the sharp identified set by extending random-set arguments to infinitely many moment restrictions. We then develop estimation and inference as the number of moments increases, using methods from the many-moment-inequality literature to control a high-dimensional support-function process.

My broader research studies robust inference when the available information does not justify conventional strong assumptions. In *Robust Bayesian Inference under Partially Specified Priors*, I replace a single fully specified prior with classes of priors satisfying economically motivated restrictions. I show that robust posterior bounds solve infinite-dimensional linear programs and develop a simulation-based approximation that reduces these problems to finite-dimensional optimization.

My other work develops practical and computationally tractable econometric methods, including adaptive information pooling in Bayesian VARs, conformal methods for quantile forecasting, and a Stata implementation of confidence-set methods for model and forecast comparison.

I believe the HIDI project would be an excellent environment for developing my research. The closest connection is to my work on partial identification with an increasing number of moment restrictions, which relates directly to Professor Kock's research on inference in partially identified models with many moment inequalities. More broadly, the project's focus on high-dimensional hypothesis testing, robust estimation and testing, and Gaussian and bootstrap approximations connects naturally to open questions arising in my work on nonregular and increasingly high-dimensional inference.

I would be delighted to join the Department of Economics at Oxford and contribute to the HIDI project. Thank you for considering my application. I look forward to the possibility of discussing my research with you.

Yours sincerely,

Martin Fankhauser